RAMKUMAR SHARMA

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PROFESSIONAL SUMMARY

AI Engineer with 1+ year of experience building production-grade Generative AI and LLM applications, currently contributing to AI-powered products at Claribel.ai. Specialized in Retrieval-Augmented Generation (RAG) pipelines, LangChain and LangGraph orchestration, Pinecone vector database integration, and semantic search systems on AWS serverless infrastructure. Proficient in Python, OpenAI API, prompt engineering, embeddings, hybrid search, reranking, LLM evaluation, and full-stack AI application development with React.js and REST APIs. Focused on scalability, cost optimization, and monitoring and observability using CloudWatch.

TECHNICAL SKILLS

AI & GenAI: LLM, Generative AI (GenAI), RAG, Prompt Engineering, Embeddings, Semantic Search, Hybrid Search, Reranking, LLM Evaluation, AI System Design, Function Calling, Retrieval Optimization

Frameworks & Libraries: LangChain, LangGraph, OpenAI API, Python 3.11+, Node.js, JavaScript

Vector Databases & Storage: Pinecone, PostgreSQL, DynamoDB, MongoDB, AWS S3

Cloud & Infrastructure: AWS Lambda, API Gateway, CloudWatch, Serverless Architecture, Docker, CI/CD (GitHub Actions), Cost Optimization, Monitoring and Observability

Frontend & APIs: React.js, REST API, JSON, Authentication, Authorization (JWT)

Developer Tools: Git, GitHub, SQL, MongoDB, Jira

PROFESSIONAL EXPERIENCE

Claribel.ai | San Francisco Bay Area, CA (Remote)

AI Engineer — Full-Stack AI Application Development *June 2025 – Present*

- Develop and maintain AI-powered product features using Python, OpenAI API, and LangChain, contributing to LLM application pipelines that serve production users with sub-3-second response latency.
- Build and integrate Retrieval-Augmented Generation (RAG) workflows using vector database search and semantic retrieval, enabling AI features to answer queries with contextually accurate, source-grounded responses.
- Implement prompt engineering strategies and structured prompt templates that improve LLM output quality, reducing hallucinated responses by an estimated 35% during internal testing cycles.
- Develop React.js frontend components interfacing with AI backend services via REST APIs, delivering responsive, production-grade AI user experiences aligned with Figma design specs.
- Integrate AWS Lambda and API Gateway to expose serverless AI endpoints, supporting scalable, cost-efficient inference pipelines without dedicated server infrastructure.
- Implement JWT-based Authentication and Authorization for AI-powered product modules, ensuring secure, role-based access to LLM features across different user tiers.
- Participate in LLM evaluation cycles — testing response accuracy, latency benchmarks, and retrieval precision — contributing to data-driven model selection decisions.
- Collaborate cross-functionally using Jira, documenting AI system design decisions and maintaining REST API integration specs for backend AI services.

AI PROJECTS

Enterprise RAG Assistant

Python | LangChain | LangGraph | Pinecone | OpenAI API | AWS Lambda | S3 | API Gateway | React.js | Docker | CloudWatch

- Designed and deployed a multi-document Enterprise RAG Assistant ingesting 500+ PDFs with automated chunking, embedding, and indexing into Pinecone, enabling semantic search across a 200K+ chunk corpus.

- Implemented hybrid search (dense + BM25 sparse) with cross-encoder reranking, achieving 91% retrieval precision and sub-2-second end-to-end response latency.
- Deployed on AWS serverless architecture (Lambda, API Gateway, S3) with Docker and CI/CD via GitHub Actions; cut infrastructure cost by 45% vs. EC2 baseline.
- Built React.js frontend with streaming responses, source citations, and DynamoDB conversation memory; CloudWatch dashboards reduced LLM API spend by 30%.

Quote Comparison AI Platform

Python | LangChain | OpenAI API | Function Calling | PostgreSQL | AWS Lambda | React.js | REST API

- Built an AI-powered document analysis platform extracting structured data from quote PDFs using LLM function calling, reducing manual comparison time from 4 hours to under 8 minutes.
- Engineered a multi-criteria AI comparison engine with prompt engineering and JSON-structured outputs; clients made decisions 5x faster than legacy processes.
- Integrated PostgreSQL for structured data persistence, achieving 99.3% uptime and 60% reduction in redundant OpenAI API calls.

Multi-Document Chat Platform

Python | LangGraph | OpenAI API | PostgreSQL | Docker | JWT Auth | CloudWatch | CI/CD

- Developed a scalable multi-document chat platform using LangGraph and Pinecone, supporting 50+ concurrent user sessions with isolated context windows and conversation memory.
- Implemented JWT-based Authentication and Authorization (RBAC) for secure multi-tenant document access.
- Established CI/CD pipeline via GitHub Actions and Docker, reducing release cycle from weekly to daily with zero-downtime deployments and automated test gates.

AI Customer Support Agent

Python | LangChain | OpenAI API | Function Calling | DynamoDB | API Gateway | AWS Lambda | Semantic Search

- Built a production AI Customer Support Agent using LangChain and function calling, automating 68% of Tier-1 ticket resolutions across a 10,000-entry knowledge base.
- Designed workflow automation pipeline for order lookup, policy retrieval, and escalation handling; cut average resolution time from 18 minutes to 3.5 minutes.
- Integrated semantic search with RAG retrieval, achieving 87% first-contact resolution and reducing human escalation volume by 52%.

EDUCATION

Master of Computer Applications (MCA) | DIT University, Roorkee | Jul 2023 – Jun 2025

Relevant Coursework: Software Engineering, Artificial Intelligence, Java, Web Technology

Bachelor of Computer Applications (BCA) | Quantum University, Roorkee | Jul 2020 – Jun 2023

Relevant Coursework: Machine Learning, Data Structures, Java, Database Management Systems

LANGUAGES

English — Intermediate | **Hindi** — Intermediate